

Kingdom of Saudi Arabia
Ministry of Higher Education
King Faisal University
College of Computer Sciences & Information Technology

Library and Bookstore Management System

Database Concepts and Design Course Project

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Serial No	Name	ID	Sec
17	Reda Alqatifi		01
11	Abdullah Alkhathlan		01
11	Ali Alburahim		05
17	Ali Alkhars		05

Instructor: Muhammad Aasim Rafique

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1. Introduction (case description)

To effectively manage and optimize its operations, the **Library** requires a robust system that consolidates and tracks crucial information about its **employees, customers, departments, and Books**. This centralized system serves as the backbone for ensuring that the **Library** operates smoothly, providing accessible and accurate data that supports decision-making processes across all levels of the organization.

By maintaining detailed records of **employee** information, **customer** information, **departmental** structures, and all **books** —including those available in the system or ordered by customers—the system enables seamless coordination and communication. It also facilitates efficient human resource management, workforce planning, and improved book management.

Therefore, a well-structured solution is pivotal for the **Library ability** to adapt and respond to dynamic business needs, ultimately driving productivity and fostering informed strategic planning.

2. System Analysis

2.1 Requirement Specification

1. The Library is organized into **departments**. Each department has a **unique name**, a **unique ID**, and a particular **employee** who **manages** the department.
2. Each **Book** in the system has a **unique code**, a **unique name**, a **price**, a **stock**. Also, every book must **contain** at least one **category** and it must be **written by** at least one **author**. Books can be **borrowed** or **purchased**, and they must be recorded.
 - a. **Borrowed books** must be **related** to the **order** of the **customer** who borrowed it, and it must have a **return date**.
 - b. **Purchased books** must be **related** to the **order** of the **customer** who purchased it.
3. The **author** has a **unique ID**, a **nick-name**, and a **unique email** to contact with him. Every author can **write** one or more books.
4. All **categories** have a **unique ID** and a **unique name**. Also, one category can be **contained by** more than one book.
5. Every **user** has a **unique ID**, a **name**, a **unique email**, a **gender**, and a **hashed password**. Also, each **user** may have a **birth date**. There are 2 types of users, which are:
 - a. The **customer** has an **address** and he can **order** one **book** or more—borrow and purchase—Each order has a **unique number**, **order date**, and a **total price**. Also, any customer can be **served by** one **employee** or more.
 - b. The **employee** has a **unique key-code** giving by his department **manager**, a **salary**, and **working hours**. Also, each employee must **work for** a **department** and this **department** must be **managed by** only one employee. Also, any **employee** can **serve** one **customer** or more. But only the **manager** can **manage** books. Also, every customer may be **dependent of** dependents.
 - i. Each **dependent** has a **unique name**, **gender**, and the **relationship** between him and the employee.

A. List of users

1. Employee (including Manager role).
2. Customer.

B. List of main functions (at least three)

1. Store and maintain detailed employee profiles, which include personal information, contact details, and work information such as salary and work hours.
2. Organize and categorize employees into various departments, enabling easy tracking of departmental structure.
3. Manage Books by the manager such as adding, deleting, updating, and ensuring clear oversight of purchased and borrowed books.

4. Generate detailed reports on various aspects, such as department details, orders details, and customer data, Load on employee.
5. Implement role-based access controls to ensure that sensitive information is accessible only to authorized personnel.
6. Manage customer orders (borrowing and purchasing), including recording return dates for borrowed books and tracking order history.
7. Store and maintain the list of dependents of employees for the family support claims or insurance claims

C. List of main reports (at least three)

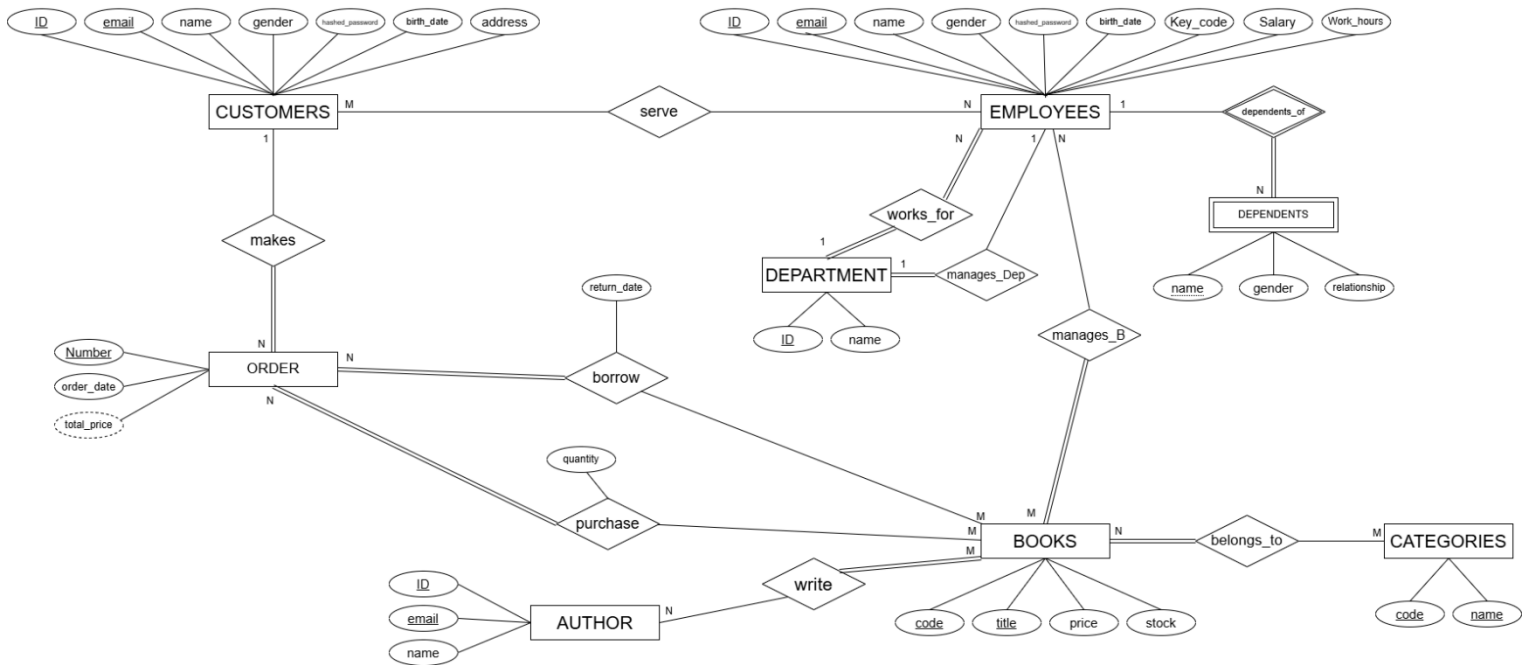
1. Books with low or zero stock in the inventory.
2. Most borrowed/purchased books by category.
3. Number of authors who published their books in the library.
4. Number of customers and employees with their details.
5. Orders details, for whom and when.
6. Department details such as its employees and manager.

3. Database Conceptual Design (ERD)

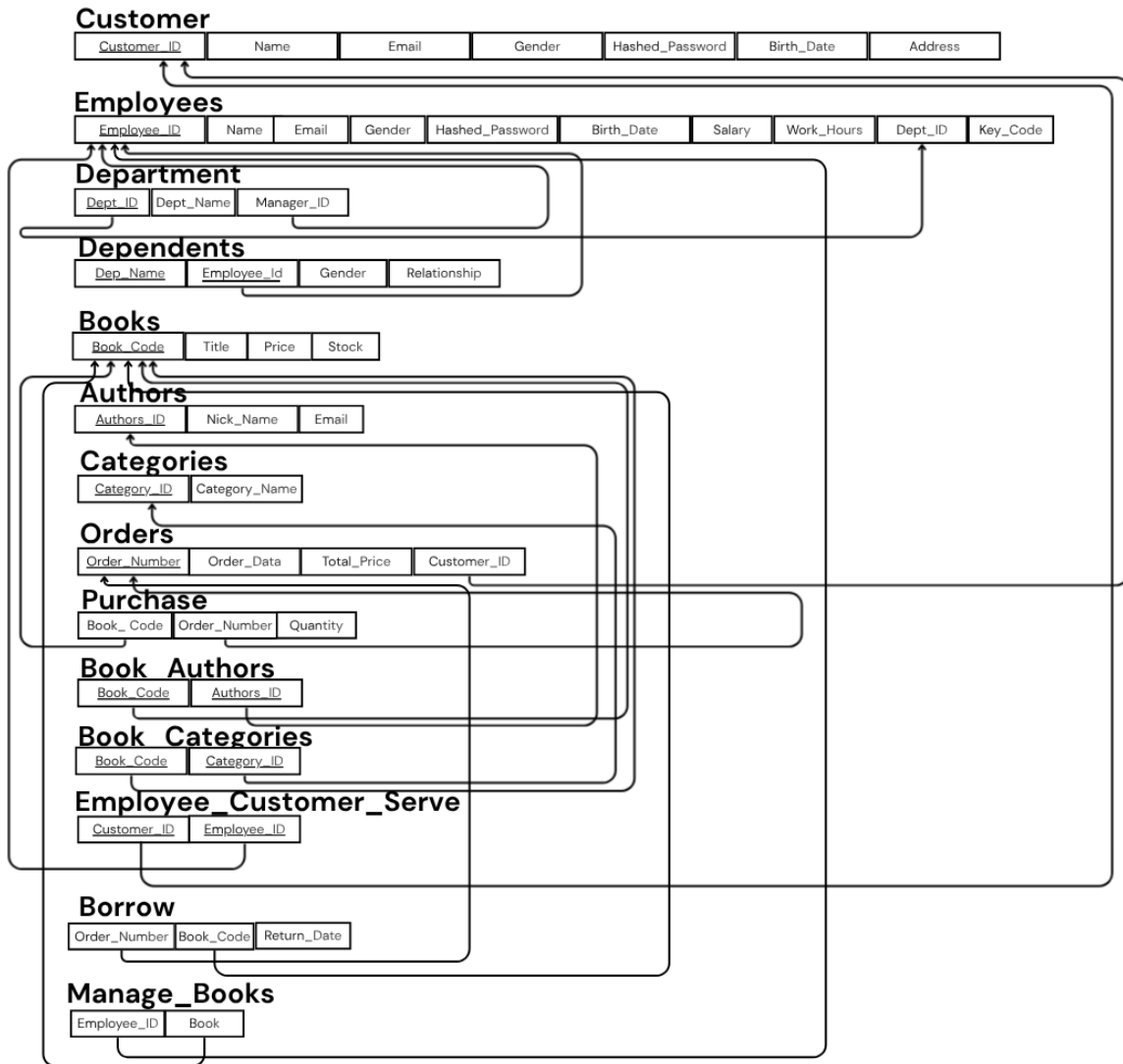
Library and Bookstore Management System

Conceptual Design (ERD)

Using: Oracle Apex



4. Logical Design (Relational Schema)



5. Physical Design

Customer	<pre>CREATE TABLE Customer (ID INT PRIMARY KEY, Name VARCHAR(30) NOT NULL, Email VARCHAR(50) UNIQUE NOT NULL, Gender CHAR(1) CHECK (Gender IN ('M', 'F')), Hashed_Password VARCHAR(255) NOT NULL, Birth_Date DATE, Address VARCHAR(50));</pre>
Department	<pre>CREATE TABLE Department (Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR (50) UNIQUE NOT NULL, Manager_ID INT);</pre>
Employees	<pre>CREATE TABLE Employees (Employee_ID INT PRIMARY KEY, Name VARCHAR(30) NOT NULL, Email VARCHAR(50) UNIQUE NOT NULL, Gender CHAR(1) CHECK (Gender IN ('M', 'F')), Hashed_Password VARCHAR(255) NOT NULL, Birth_Date DATE, Salary DECIMAL(10,2) NOT NULL, Work_Hours INT NOT NULL, Dept_ID INT, Key_Code VARCHAR(20) UNIQUE NOT NULL, FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID));</pre>
Alter the table "Department"	<pre>ALTER TABLE Department ADD CONSTRAINT FK_Manager FOREIGN KEY (Manager_ID) REFERENCES Employees(Employee_ID);</pre>

Dependents	<pre> CREATE TABLE Dependents (Dep_Name VARCHAR(30), Employee_ID INT NOT NULL, Gender CHAR(1) CHECK (Gender IN ('M', 'F')), Relationship VARCHAR(20) NOT NULL, PRIMARY KEY (Employee_ID, Dep_Name), FOREIGN KEY (Employee_ID) REFERENCES Employees(Employee_ID)); </pre>
Books	<pre> CREATE TABLE Books (Book_Code INT PRIMARY KEY, Title VARCHAR(100) NOT NULL, Price DECIMAL(10,2) NOT NULL, Stock INT NOT NULL); </pre>
Authors	<pre> CREATE TABLE Authors (Authors_ID INT PRIMARY KEY, Nick_Name VARCHAR(30) NOT NULL, Email VARCHAR(50) UNIQUE NOT NULL); </pre>
Catagories	<pre> CREATE TABLE Categories (Category_ID INT PRIMARY KEY, Category_Name VARCHAR(50) UNIQUE NOT NULL); </pre>
Orders	<pre> CREATE TABLE Orders (Order_Number INT PRIMARY KEY, Order_Date DATE NOT NULL, Total_Price DECIMAL(10, 2) NOT NULL, Customer_ID INT NOT NULL, FOREIGN KEY (Customer_ID) REFERENCES Customer(ID)); </pre>
Purchase	<pre> CREATE TABLE Purchase (Book_Code INT, Order_Number INT, Quantity INT NOT NULL, PRIMARY KEY (Book_Code, Order_Number), FOREIGN KEY (Book_Code) REFERENCES Books(Book_Code), FOREIGN KEY (Order_Number) REFERENCES Orders(Order_Number)); </pre>
Borrow	<pre> CREATE TABLE Borrow (Order_Number INT, </pre>

	Book_Code INT, Return_Date DATE NOT NULL, PRIMARY KEY (Order_Number, Book_Code), FOREIGN KEY (Order_Number) REFERENCES Orders(Order_Number), FOREIGN KEY (Book_Code) REFERENCES Books(Book_Code));
Book_Authors	CREATE TABLE Book_Authors (Book_Code INT, Authors_ID INT, PRIMARY KEY (Book_Code, Authors_ID), FOREIGN KEY (Book_Code) REFERENCES Books(Book_Code), FOREIGN KEY (Authors_ID) REFERENCES Authors(Authors_ID));
Book_Categories	CREATE TABLE Book_Categories (Book_Code INT, Category_ID INT, PRIMARY KEY (Book_Code, Category_ID), FOREIGN KEY (Book_Code) REFERENCES Books(Book_Code), FOREIGN KEY (Category_ID) REFERENCES Categories(Category_ID));
Manage_Books	CREATE TABLE Manage_Books (Employee_ID INT, Book INT, PRIMARY KEY (Employee_ID, Book), FOREIGN KEY (Employee_ID) REFERENCES Employees(Employee_ID), FOREIGN KEY (Book) REFERENCES Books(Book_Code));
Employee_Customer_Serve	CREATE TABLE Employee_Customer_Serve (Customer_ID INT, Employee_ID INT, PRIMARY KEY (Customer_ID, Employee_ID), FOREIGN KEY (Customer_ID) REFERENCES Customer(ID), FOREIGN KEY (Employee_ID) REFERENCES Employees(Employee_ID));

6. Populate Database

Populate each table of your database with enough data (records) to answers certain queries.

7. Queries

You should run a number of test queries to see that you have loaded your database in the way you intended. Run at least five test queries to show interesting results.

e.g.

Generate a list of students in section 61 who took the database with Dr. Shaheen.

```
SELECT St_name
```

```
FROM STUDENT, SECTION, COURSE, INSTRUCTOR
```

```
WHERE Sec_no = 61 AND C_Name= "Database" AND I_Name =" Dr. Shaheen"
```

SELECT Book_Code, Title, Price, Stock FROM Books WHERE Stock <= 5 ORDER BY Stock ASC;	-- 1. Books with low or zero stock in the inventory -- (Assuming low stock is 5 or less)
SELECT c.Category_Name, b.Title, (SELECT COUNT(*) FROM Borrow br WHERE br.Book_Code = b.Book_Code) + (SELECT COALESCE(SUM(p.Quantity), 0) FROM Purchase p WHERE p.Book_Code = b.Book_Code) AS Total_Transactions FROM Books b JOIN Book_Categories bc ON b.Book_Code = bc.Book_Code JOIN Categories c ON bc.Category_ID = c.Category_ID ORDER BY c.Category_Name, Total_Transactions DESC;	-- 2. Most borrowed/purchased books by category
SELECT COUNT(DISTINCT Authors_ID) AS Published_Authors_Count FROM Book_Authors;	-- 3. Number of authors who published their books in the library
SELECT 'Customers' AS User_Type, COUNT(*) AS Total_Count FROM Customer UNION SELECT 'Employees' AS User_Type, COUNT(*) AS Total_Count FROM Employees; B : SELECT ID, Name, Email, Gender, Birth_Date, Address FROM Customer;	-- 4. Number of customers and employees -- Query A: Get the total count of both customers and employees -- Query B: Display customer details -- Query C: Display employee details

<i>C :</i> <i>SELECT Employee_ID, Name, Email, Salary, Work_Hours, Dept_ID FROM Employees;</i>	
<i>SELECT o.Order_Number, o.Order_Date, o.Total_Price, c.Name AS Customer_Name, c.Email</i> <i>FROM Orders o</i> <i>JOIN Customer c ON o.Customer_ID = c.ID</i> <i>ORDER BY o.Order_Date DESC;</i>	-- 5. Orders details, for whom and when
<i>SELECT d.Dept_Name, m.Name AS Manager_Name, e.Name AS Employee_Name, e.Employee_ID, e.Work_Hours</i> <i>FROM Department d</i> <i>LEFT JOIN Employees m ON d.Manager_ID = m.Employee_ID</i> <i>LEFT JOIN Employees e ON d.Dept_ID = e.Dept_ID</i> <i>ORDER BY d.Dept_Name;</i>	-- 6. Department details such as its employees and manager

8. Application Development

Develop the application which will be used by users to access data and perform the certain business operation. The application should have the following minimum interfaces:

Application Guidelines

1. ERD – No entity types are isolated, all connected
2. ERD Mapping to Tables – All tables are correctly mapped from ERD
3. All tables correctly implemented (at least 1 table for each student) - it will include Add, Edit, delete and searching.
4. Relationship between tables working.
5. proper addition of constraints at least – PK and FK
6. Proper Data is inserted in the database
7. Every table should have main **FORM** where we can perform the following tasks:
 - a. Add new data to tables using forms. Apply validation on each field (e.g. if Primary key is duplicate It should display an error message)
 - b. Delete/ update record from each table
 - c. If a record which refers to another table is deleted or updated; apply CASCADE Update and Cascade Delete (Primary foreign key relationship)
 - d. Navigation between records (e.g. MoveFirst, MoveLast, Next, Previous etc..)
8. Implement each business functions for different users listed under sect 2-B
9. Implement all reports which you have listed under 2-C, where the user should able to select few parameters and the system should provide a result summary.

You should provide screenshots of the application in your report with a proper description of each functionality in the order it has been performed. Each figure should be labeled and cited in the text.

10. General Guidelines

Report Guidelines

- Create Table Queries written and shown in report
- Show screenshots of proper addition of constraints at least – PK and FK in report
- Proper Forms shown for input on report
- Screenshots shown with data on report
- Report Comprehension and Accuracy (Table of Contents, Introduction, Description of ERD, Description of Tables, Figures captions, tables captions are there and introduced in text)

Presentation Guidelines

- Give project demo with good presentation skills.
 - If any group have changed the titles of the tables in the application. The application is properly configured and demonstrated with good titles.
-

Tasks to do:

No	Task	Student	Dou date	Status
T1	Problem Statement	Reda	31 th March	☒
T2	SRS	Reda	31 th March	☒
T3	Conceptual Design (ERD)	Reda	12 th April	☒
T4	Logical Model	Alburahim	1 th May	☒
T5	Population of Data	Alburahim	10 th May	
T6	SQL Queries for Retrial of data for report (Complexity like using joins etc.)	Alburahim		☒
T7	Forms	Abdullah		
T8	Reports	Alkhars		
T9	Physical Design (code)	Alkhars (6 tables)		☒
		Abdullah (6 tables)		☒
		Alburahim (1 table)		☒
		Reda (1 table)		☒